Data Technician Workbook Week 1 Excel

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Table of contents

Day 1: Task 1 3

Day 2: Task 1 4

Day 2: Task 2 5

Day 3: Task 1 6

Day 3: Task 2 7

Dataset: 7

Step 1: Create a Pivot Table 7

Step 2: Use the SWITCH Function 7

Submission: 8

Day 3: Task 3 9

Course Notes 10

Additional Resources 11

Additional formation 12

Day 1: Task 1

Please research and complete the below boxes on common laws and regulations that must be followed when working with customers data, use the below bulleted list to support your answers.

- What is it
- Why is it important
- Provide a real-world example of how you can follow it
- How does it impact working with data
- What could happen if you breached it

Data Protection Act	**What is it?** The Data Protection Act 2018 is a UK Law that governs how personal data is processed and works alongside the UK GDPR. **Why is it important?** It provides legal protection for people's personal data information and sets requirements for organisations that collect and use data. **Example:** When working with customer records, you should only access information you need for your job, keeping it confidential and ensuring all records both paper and electronic are stored securely. **How does it impact working with data?** It means organisations and employees must handle personal information responsibly and carefully, following the DPA requirements when collecting, storing, using or sharing it. **What would happen if breached?** This could result in reinforcement action, financial penalties, legal consequences and damage to an organisation's reputation. Employees could also face disciplinary action depending on the circumstances.
GDPR	**What is it?** A set of rules that controls how organisations collect, use, store and share people's personal data. This gives individuals the right to choose how their personal information is used or handled. **Why is it important?** It helps protect people's privacy and ensures that organisations handle personal data fairly, securely and responsibly. **Example:** Only collecting the customer information that is necessary for the service being provided, storing it securely, and sharing it with authorised people only. **How does it impact working with data?** It means organisations need to have a lawful reason for processing personal data, keep data accurate and secure, whilst following the rules on handling the data. **What would happen if breached?** This could result in enforcement action and significant financial penalties. It could also suffer reputational damage and loss of customer trust. Individuals may also have rights to complain or seek compensation in certain circumstances.

Freedom of Information Act	**What is it?** This law gives people the right to request record information held by public authorities, such as government departments, councils and NHS organisations. **Why is it important?** It promotes transparency and accountability by allowing the public to find out how public organisations operate and how decisions are made. **Example:** If you work for an NHS organisation and receive a freedom of information request, you should follow the organisation's FOI procedures and pass the request to the appropriate FOI/data protection team rather than responding yourself. **How does it impact working with data?** Employees need to understand that some information held by their organisation may need to be disclosed when a valid FOI request is made. However, personal or confidential information is no automatically disclosed and exemptions may apply. **What would happen if breached?** This could result in enforcement action for failing to comply with its legal FOI obligations. Employees could also face disciplinary action depending on the circumstances. Incorrectly disclosing confidential or personal information could also create data protection issues, complaints and reputational damage
Computer Misuse Act	**What is it?** Law that makes certain types of unauthorised access to or interference with computer systems illegal. **Why is it important?** It helps protect computer systems and the information stored on them from unauthorised access, hacking and other forms of misuse. **Example:** Only use customer databases and systems that you have been authorised to access. Never use another employee's login details or attempt to access information that you do not need for your role. **How does it impact working with data?** It means employees must use computer systems responsibly and only access data and systems that have permission to use. **What would happen if breached?** Depending on the offence, a person could face disciplinary action and criminal prosecution, including fines or imprisonment. An organisation could also suffer data loss, financial damage and reputational harm.

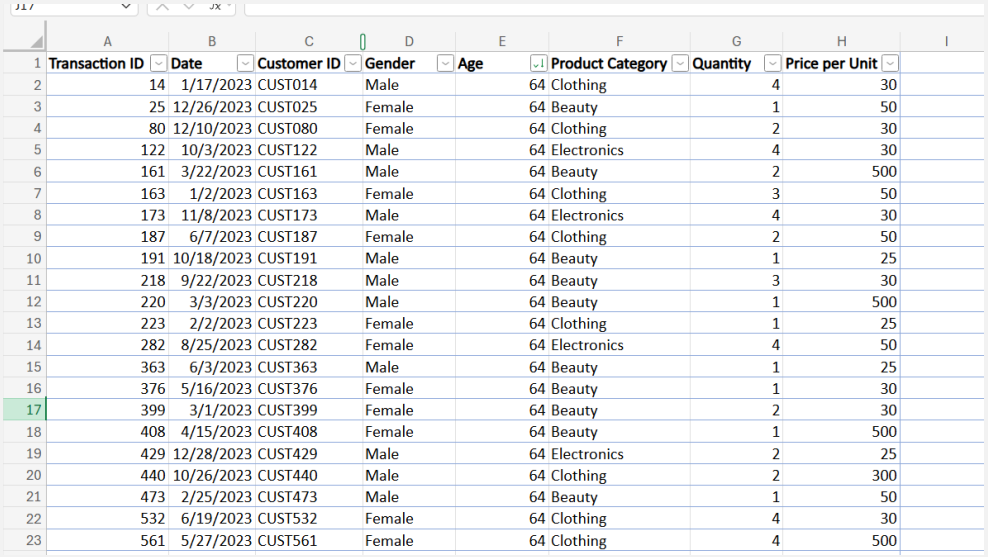
Day 2: Task 1

Please research and complete the following tasks within
the retail-sales_dataset.xlsx document, paste a print screen into the provided boxes below:

1. Using the 'filter' function, filter 'Age' to 'largest to smallest'. Sort results by Transaction ID to revert to original sorting order.
2. Using the 'SUM' function, show me the commission total in cell '**P10'**
3. Using the 'AVERAGE' function, show me the average commission in cell **'P11'**

Print screen 1

Filtering age:



Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit
14	1/17/2023	CUST014	Male	64	Clothing	4	30
25	12/26/2023	CUST025	Female	64	Beauty	1	50
80	12/10/2023	CUST080	Female	64	Clothing	2	30
122	10/3/2023	CUST122	Male	64	Electronics	4	30
161	3/22/2023	CUST161	Male	64	Beauty	2	500
163	1/2/2023	CUST163	Female	64	Clothing	3	50
173	11/8/2023	CUST173	Male	64	Electronics	4	30
187	6/7/2023	CUST187	Female	64	Clothing	2	50
191	10/18/2023	CUST191	Male	64	Beauty	1	25
218	9/22/2023	CUST218	Male	64	Beauty	3	30
220	3/3/2023	CUST220	Male	64	Beauty	1	500
223	2/2/2023	CUST223	Female	64	Clothing	1	25
282	8/25/2023	CUST282	Female	64	Electronics	4	50
363	6/3/2023	CUST363	Male	64	Beauty	1	25
376	5/16/2023	CUST376	Female	64	Beauty	1	30
399	3/1/2023	CUST399	Female	64	Beauty	2	30
408	4/15/2023	CUST408	Female	64	Beauty	1	500
429	12/28/2023	CUST429	Male	64	Electronics	2	25
440	10/26/2023	CUST440	Male	64	Clothing	2	300
473	2/25/2023	CUST473	Male	64	Beauty	1	50
532	6/19/2023	CUST532	Female	64	Clothing	4	30
561	5/27/2023	CUST561	Female	64	Clothing	4	500

Print screen 2

2023		
commision total:	£	146.33
Average:	£	6.65
2024		
commision total:	£	195.10
Average:	£	8.87

Print screen 3

	Total Sales	Quantity	
All products		female	Male
Beauty	52090	418	353
Electronics	0	439	410
clothing	50830	441	453

Day 2: Task 2

Using the skills developed today, have some fun with the data set you have imported. Paste your work below and enjoy!

Print screen 1

Calculate the average for all students and fill into Column E. (Use formula)

Student name	English	Mathema	Science	Average
Carol	75	85	85	81.666667
John	65	80	70	71.666667
Edward	55	80	60	65
Khan	85	75	80	80
Ted	80	75	90	81.666667
Harry	80	70	80	76.666667
Sarah	80	70	80	76.666667
Mary	55	70	65	63.333333
Linda	90	50	70	70
Thomas	55	30	65	50

Print screen 2

Using the =MAX fucntion, tell me what the students highest score was in column F.

Student name	English	Mathema	Science	Average	Highest sco
Carol	75	85	85	81.666667	85
John	65	80	70	71.666667	80
Edward	55	80	60	65	80
Khan	85	75	80	80	85
Ted	80	75	90	81.666667	90
Harry	80	70	80	76.666667	80
Sarah	80	70	80	76.666667	80
Mary	55	70	65	63.333333	70
Linda	90	50	70	70	90
Thomas	55	30	65	50	65

Print screen 3

Apply filter and sorting to show the best student in this classroom by average.

Student name	English	Mathema	Scienc	Averag	Highest sco
Carol	75	85	85	81.666667	85
Ted	80	75	90	81.666667	90
Khan	85	75	80	80	85
Harry	80	70	80	76.666667	80
Sarah	80	70	80	76.666667	80
John	65	80	70	71.666667	80
Linda	90	50	70	70	90
Edward	55	80	60	65	80
Mary	55	70	65	63.333333	70
Thomas	55	30	65	50	65

Apply filter and sorting to show the best student in this classroom by highest score.

Student name	English	Mathema	Scienc	Averag	Highest sco
Ted	80	75	90	81.666667	90
Linda	90	50	70	70	90
Carol	75	85	85	81.666667	85
Khan	85	75	80	80	85
Harry	80	70	80	76.666667	80
Sarah	80	70	80	76.666667	80
John	65	80	70	71.666667	80
Edward	55	80	60	65	80
Mary	55	70	65	63.333333	70
Thomas	55	30	65	50	65

Print screen 4

Use conditional formatting to clearly identify the highest and lowest average scores.

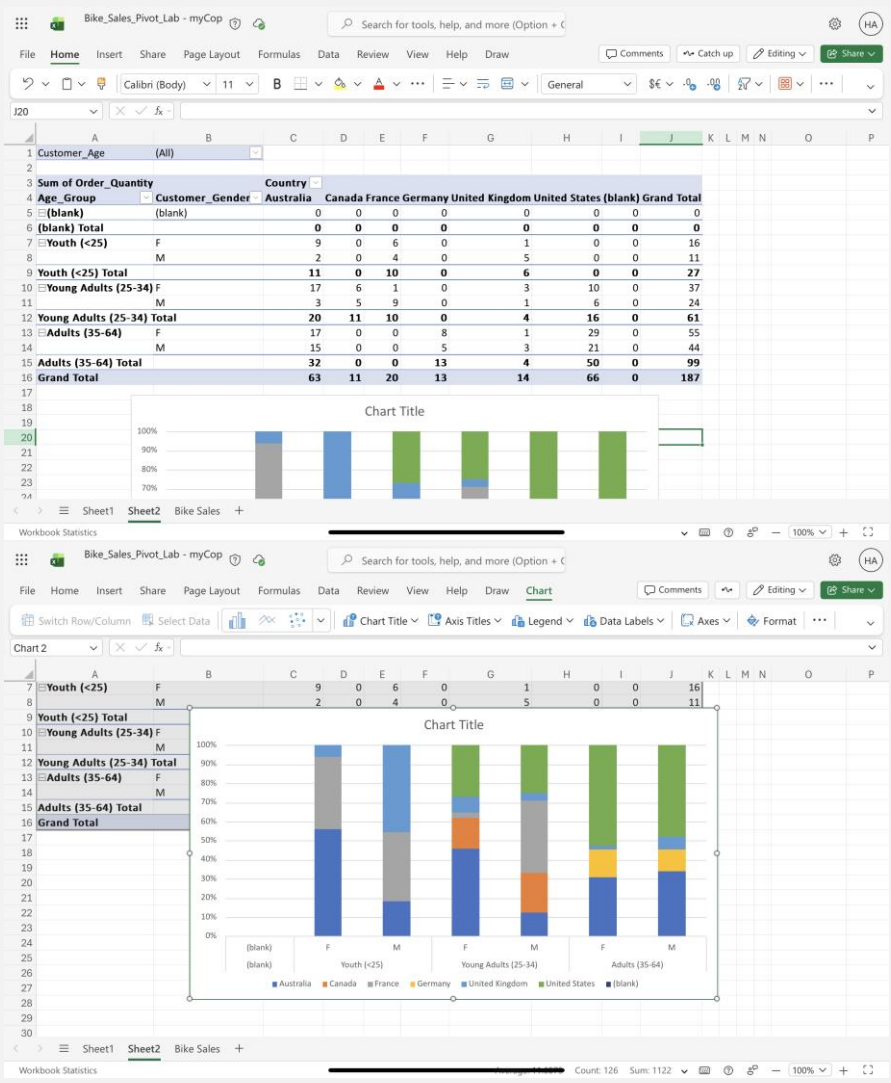
Student name	English	Mathema	Scienc	Averag	Highest sco
Ted	80	75	90	81.666667	90
Linda	90	50	70	70	90
Carol	75	85	85	81.666667	85
Khan	85	75	80	80	85
Harry	80	70	80	76.666667	80
Sarah	80	70	80	76.666667	80
John	65	80	70	71.666667	80
Edward	55	80	60	65	80
Mary	55	70	65	63.333333	70
Thomas	55	30	65	50	65

Day 3: Task 1

Please open and make an online copy of the dataset 'Bike_Sales_Pivot_Lab.xlsx from here.

The lab instructions can be found here Bike Sales Pivot Tables.docx . Do not worry if you do not complete the lab, just working with data and playing with the pivot table will be good experience.

Please paste your final pivot table below and complete the reflection questions:

Print screen 1	
In which Age Group does Germany have customers?	Adults 35-64

What country has sales in all markets?	Australia and United Kingdom
What is the most profitable country by age group, and gender?	Australia
Any other findings?	- There seems to be more females compared to males in general/total, across all age groups with a difference of 29. - Although, Australia and Uk had the highest sales the male purchase were lower for them, especially Australia, compared to all countries in general. - Australia and UK were the only ones with consistent sales, with sales in every age range and gender. - Germany had the least with it only being in the older range. - Out of all the age groups the older range, 35-64, had the most sales, suggesting that this range prefer and do bike mostly compared to the younger generation. This may be because they are too young and require picking and dropping, or because they have a car, hence new drivers or workers that are unable to bike. This older range, some may be retired, working from home or able to/ enjoy a bike ride.

Day 3: Task 2

The dataset below tracks the sales performance of different products in various counties in England. Please paste the dataset into a blank Excel workbook. Your task is to:

- **Create a Pivot Table** to summarise the data by county and product.
- **Use the SWITCH function** to categorise products based on their sales volume.

Dataset:

County	Product	Sales Volume
Yorkshire	Laptops	500
Yorkshire	Smartphones	200
Cornwall	Laptops	700
Cornwall	Printers	400
Lancashire	Smartphones	150
Lancashire	Laptops	600
Essex	Printers	800
Essex	Smartphones	300
Durham	Laptops	250
Durham	Printers	300
Greater Manchester	Smartphones	600
Greater Manchester	Laptops	400

Step 1: Create a Pivot Table

- Select the dataset (columns A to C).
- Insert a Pivot Table to summarise the data by **County** in the rows and **Products** in the columns. Use **Sales Volume** as the value to be summarised.

Step 2: Use the SWITCH Function

In a new column next to your data, use the SWITCH function to categorise products based on **Sales Volume** as follows:

- For sales greater than 600: **"High"**
- For sales between 300 and 600: **"Medium"**
- For sales less than 300: **"Low"**

SWITCH Function Example:

```
=SWITCH(TRUE, C2 > 600, "High", C2 >= 300, "Medium", "Low")
```

- Apply this formula to each row and check if the products are categorised correctly.

Submission:

- A completed Pivot Table summarising sales by county and product.
- A new column in the dataset categorising products by sales volume using the SWITCH function.
    - Please paste your completed work below

Print screen 1

Count of Sales Volume	Product					
County	Laptops	Printers	Smartphones	(blank)	Grand Total	
Cornwall	1	1			2	Low
Durham	1	1			2	Low
Essex		1	1		2	High
Greater Manchester	1		1		2	Low
Lancashire	1		1		2	Low
Yorkshire	1		1		2	Low
(blank)						Low
Grand Total	**5**	**3**	**4**		**12**	Low

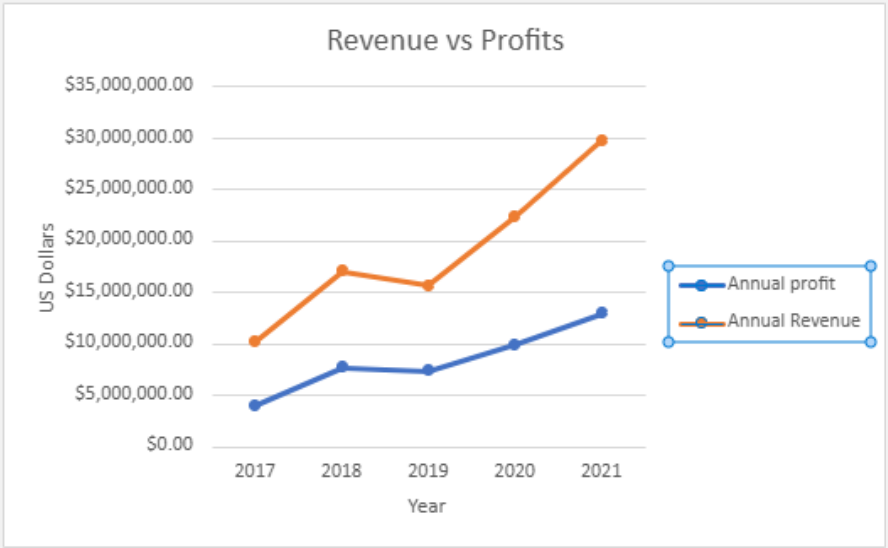
Day 3: Task 3

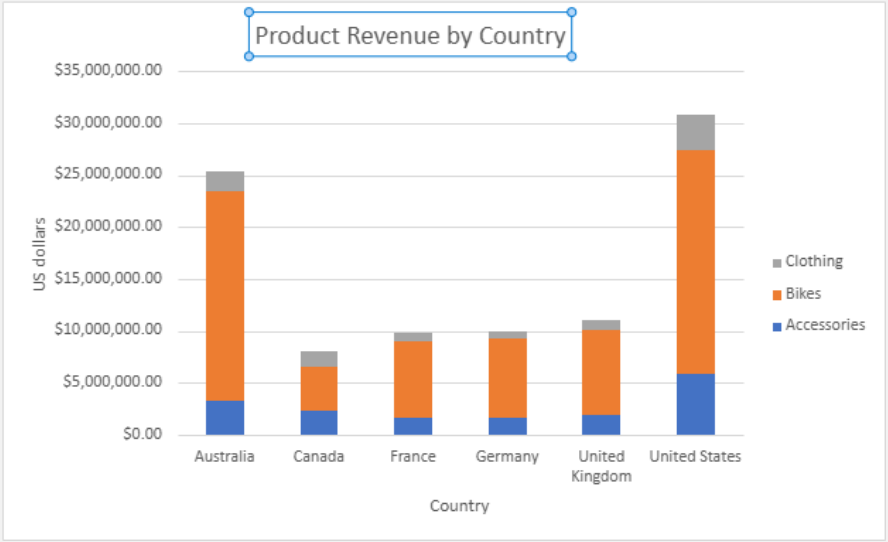
Please open and make an online copy of the dataset 'Bike_Sales_Visualizations_Lab.xlsx from here.

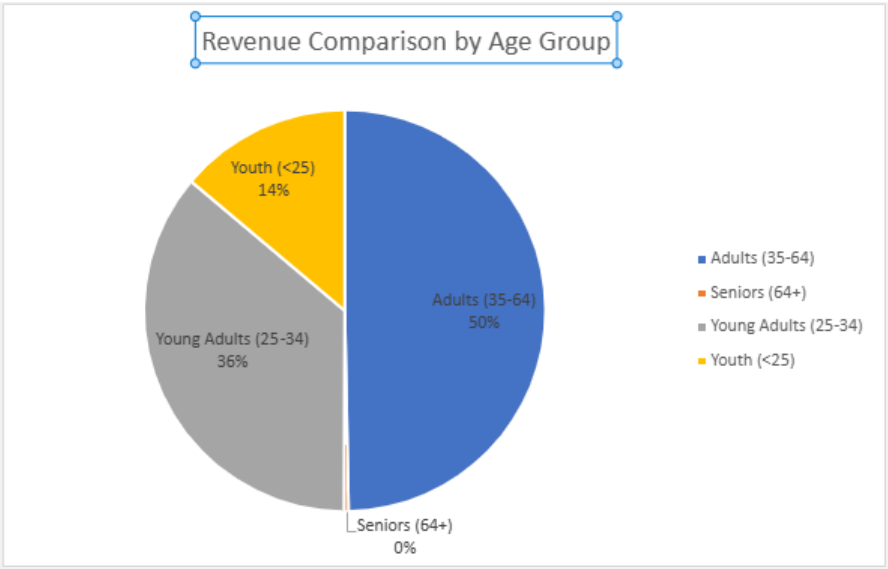
The lab instructions can be found here Bike Sales Visualization Lab.docx. Do not worry if you do not complete the lab, just working with data and playing with the charts will be good experience.

Please paste your results below:

Print screen 1






Revenue Comparison by Age Group
Youth (<25)
14%
Adults (35-64)
50%
Young Adults (25-34)
36%
Seniors (64+)
0%
Adults (35-64)
Seniors (64+)
Young Adults (25-34)
Youth (<25)

Course Notes

It is recommended to take notes from the course, use the space below to do so, or use the revision guide shared with the class:

Data analysis = The process of inspecting, cleansing, transforming and modelling sata to discover useful information and support decision-making.
Types of data classification: based on... Structure, Format (Data Type) and statistical meaning.

A text string is a sequence of characters used in computing to represent textual data. Essential for handling & manipulating text-based info.
Numeric represents numbers, inc intergers, floating-point numbers and complex numbers.
Logical/Boolean- a data type with only two possible values: true or false, representing truth values. Used to control program flow, make decision and manage state conditions in software applications.

Quantitative discrete: variable is a type that represents countable data that can only take on specific, distinct values. These values are typically numbers (0,10,45,34) and cannot be subdivided further.
Quantitative continuous: Variable is a type of variable that represents measureable data that can take on any value within a certain range of interval. Countinuous variables take on an infinite number of values within their defined range.
Qualitative Binomal: Data refers to categorial data that can be classified into two mutally exclusive categories of outcomes. The term 'binomal' indicates that there are only two possible outcomes. These outcomes are typically non-numeric and represent different categories or characteristics.
Qualitative Ordinal: data refers to categorial data where the categories have a natural order or ranking. The intervals between the caegories are not necessarily equal or meaningful.
Quantitative normal: data refers to categorial data that represents different categories or groups with no inherent order of ranking among them. These categories are typically lables or names that classify items into distinct groups based on some characteristic.

- In the *Data* menu, click on the *Subtotal* option under the *Outline* section so that a configuration window appears.
- Select the column `CustomerCountry` in the *At each change in:* dropdown menu.
- Select *Average* as the aggregation method in the *Use function:* dropdown menu.
- Ensure only `OrderToDelivery` is checked in the list of columns in the *Add subtotal to:* section.
- If not already ticked, tick the *Summary below data* option.

Arithmetic operators:
Addition (=A1+B1)
Multiplication (=A1*B1)
Division (=A1/B1)
Subtraction (=A1-B1)
Percentage (=50%) or (A1*50%)...etc

SUM: Calculate the total sales value
SUMIF: Calculate the total sales value for a specific category e.g: clothing
UNIQUE(): Returns a list of unique values from a range or array
AVERAGE(): Calculate the mean of a group
AVERAGEIF(): Calculate the average of numbers that meet a specified criterion
YEAR():... MONTH():....DAY():.... Return the "..." from a date value
IF(): Logical function used for making decisions based on specific critera
IFS(): for making decision on multiple criteria

F(x)
=name(argument1,argument2,argument3) --> same/similar for other softwheres: PalBI, SQL...etc
=Vlook(what,where,return,app/exact) --> Example: =vlookup(nhs,number,table(info), firstname,false)

=Sum(1,2,3...)

- ‘False’ means find and return exact match. - ‘True’ means exact match

In rare case: if there are two of the same ID’s for example two ID11 or something, because the value you are looking for, if you put down false, this ID, the Vlookup function will go to the first ID, match the first ID and then return the piece of data. This is because there is another ID11, it will skip it, this is why we use the false statement here. Basically “Go ahead and find me the exact match of the ID11”.

=Switch(value,match,result) --> a logical function, and then you have an expression to compare (so your looking for an expression).
For example 1: your looking up the value and you want to find a match then return the result once found. Example 2: the grading boundaries in an exam, if you get 90 and high you get A+, if between 85-89 it A... etc. These type of grading boundaries can be calculated using the SWITCH function. So if you have an excel sheet where you have to replace the numbers, you are going to do it with the A+, A and A-, you can use the SWITCH function. Then if its matched to 90 it’ll put A+, 80-89 it’ll put A... and so on.

Setting a password on EXCEL APP (doesn’t work online):

Protecting the whole file/workbook: Excel desktop --> open file --> go to file --> info --> protect the workbook (dropdown) --> (choose) password.
Protecting the worksheet only:

- So people need password to edit or view depending on password type, invite link who you send it to.

CONCENTRATE: combines the value of cells.... =CONCATENATE(B2,” “,C2)
VLOOKUP: to find the product category for transaction of different IDs....
COUNT: function counts the number of cells in a range that contain numbers
COUNTA: functions counts the number of cells in a range that are not empty.

Conditional formatting: allows you to apply specific formatting to cells that meet certain criteria

I HAVE WRITTEN MORE THING IN MY NOTEBOOK BY HAND, JUST AS ITS EASIER AND FASTER FOR ME IN TERMS OF NOTES, I CAN SEND THEM OVER IF YOU WOULD LIKE.

Additional Resources

You can explore the additional resources to support your learning.

1) DataCamp recommended courses.
    a. Understanding GDPR - DataCamp Learn
    b. Introduction to Excel - DataCamp Learn
    c. Data Analysis in Excel - DataCamp Learn
    d. Data Preparation in Excel - DataCamp Learn
    e. Data Visualization in Excel - DataCamp Learn

2) Introduction to Data Analytics Essentials
   This data analytics essentials course teaches you the fundamental tools of a data analyst. You will learn to transform, organize, and visualize data with spreadsheet tools such as Excel.

3) Excel Formulas and Functions Tutorial
   Formulas can be used to perform simple math using +,-,*, or / to add, subtract, multiply or divide. Excel also comes built-in with over 300 functions that can be used to perform complex math, lookup values, work with text, and much more. This section will cover creating simple formulas, cell references, strings of text, and keyboard shortcuts.

4) Excel Exercises Dashboard
   Our bite-sized, interactive lessons make Excel skills easy to learn and remember.

5) 100+ Excel Interview Questions & Answers To Make You Job Ready
   This guide can also be used by interviewers who want to gauge the Excel skill of the candidates in the interview.

6) ☒ How to use Excel - YouTube
   In these step-by-step tutorials, learn how to use Microsoft Excel. Find out how to get started with MS Excel in under 10 minutes. Learn about Pivot Tables, formulas and functions, vlookup, shortcuts. Also, find out about the latest new features and the top tips and tricks.

7) Use Excel skills (**Filter, Sort, Conditional Formatting, Functions, VLOOKUP, Pivot Tables, Pivot Charts, and Slicers**) to analyse global sales data and uncover key insights.
   - Find exercises here Global Superstore Data Challenge.docx
   - Open dataset global_superstore_2016.xlsx in Excel online and create an online copy.

8) In this free Excel tutorial, learn how to create formulas and charts, use functions, format cells, and do more with your spreadsheets.
   Free Excel Tutorial at GCFGlobal

Additional formation

We have included a range of additional links to further resources and information that you may find useful, these can be found within your revision guide.

END OF WORKBOOK

Please check through your work thoroughly before submitting and update the table of contents if required.

Please send your completed work booklet to your trainer by submitting in MS Teams Assignment page.